REVEAL: Supplementary Material

Anonymous CVPR submission

A. Overview

This supplement provides runtime and training details, followed by Q3 target-search evidence, Q2 controller results and Q1 system comparisons. Q2 uses exactly three evaluation seeds, 187102, 187103 and 187104, with 40 trials per method and seed. Other seeds identify separate studies or training diagnostics. All reported rates retain their study-specific denominators. Physical-hand evaluation remains pending.

B. Task and Runtime

This section describes the cube system. The die, additional-object and language sections explain the differences for other objects.

B.1. Object and goals

Table S1 records the fixed face mapping. Body normals include the original mesh rotation. All methods use the same mapping and cached six-vector bank. The cyan channel corresponds visually to a pale-blue face and magenta to a purple face. The target quaternion gives a canonical twist as well as the upward face normal.

Letter	Skill color	Body normal	Target XYZW
T	red	$(0, -1, 0)$	$(-s, 0, 0, s)$
E	green	$(-1, 0, 0)$	$(0, s, 0, s)$
R	blue	$(0, 0, -1)$	$(1, 0, 0, 0)$
L	yellow	$(0, 0, 1)$	$(0, 0, 0, 1)$
S	magenta	$(0, 1, 0)$	$(s, 0, 0, s)$
H	cyan	$(1, 0, 0)$	$(0, -s, 0, s)$

Table S1. Frozen goal mapping, with $s = \sqrt{1/2}$.

The instruction names a letter, and the benchmark checks its fixed reference orientation with a 0.16-radian tolerance. A drop ends the trial when the object moves at least 0.24 metres from its nominal grasp position. This rule does not detect every brief slip or loss of contact within that boundary.

B.2. Physics and perception

The runtime uses Isaac Lab 3.0, Isaac Sim 6.2, Newton 1.6.1, Warp 1.17, and PyTorch 2.11 with CUDA 13.0. The validated collision model contains 23 bodies and 117 shapes, comprising 116 convex shapes and one box. The contact allocation uses `nconmax=4096` and `njmax=1024`. Earlier overflow runs are invalid even if their reported rewards appear plausible.

The pose estimator is frozen. It predicts 27 image-coordinate/depth outputs for nine points, then recovers pose through calibrated geometry. The object image is rendered at 120×120 with live hand occlusion. Sensor calibration, object asset, policy normalization, joint order, action mapping, and smoothing belong to the release contract. A change to one of these produces a new release. The current setup does not establish real-camera robustness or unseen-scene generalization.

B.3. Timing, termination, and exclusions

Each episode lasts at most 120 seconds of simulated control time. A model call taking L wall-clock seconds must remain pending for at least $\lceil 30L \rceil$ control steps, and its reply cannot be used before the call finishes. Physics continues throughout. Since simulation runs slower than real time, an episode takes longer in wall-clock time. Hardware control speed must be measured separately.

The stopping check runs at 3 Hz. The requested symbol must be bound to a goal, with no inspection or model request underway, and estimated orientation error must be at most 0.16 radians. The independent scorer then checks the actual requested face and true orientation using the same tolerance. It scores the stop time, without an extra holding period or credit for a better pose reached earlier.

A service error or model call exceeding 120 seconds is recorded without replacement. The original three-system panel removes trial 183102011 from every arm. This leaves 39 common eligible trials for that historical Q1 panel. The current Q2 study reports all 120 outcomes per method. Raw 40-trial outcomes remain visible, including service-affected cases.

C. Models and Training

Figure S1 expands the main-text task overview with the training and deployment pathways.

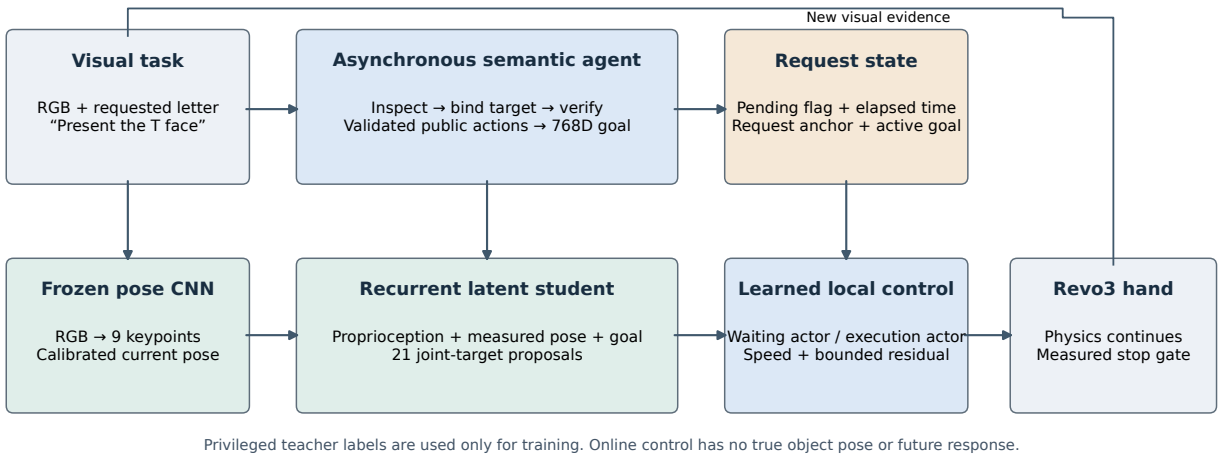


Figure S1. **Expanded system architecture.** The diagram shows privileged teacher supervision, the semantic-to-latent student interface, and closed-loop deployment. This schematic describes the cube implementation and does not report hardware outcomes.

C.1. ViserDex-based Revo3 foundation and student adaptation

Our motor foundation adapts the teacher–student approach of ViserDex [1] to the Revo3 hand. We start REVEAL adaptation from the resulting pretrained teacher and student. The teacher is frozen while the student acquires its goal-bank interface. Compatible observation encoders, recurrent and actor weights, and normalization statistics are reused. The new interface removes the target quaternion and relative-goal quaternion from the student's deployed inputs.

The student is fitted on collected trajectories using frozen-teacher action labels. Goal-projection warmup uses action MSE. Recurrent adaptation additionally uses a training-only decoder with normalized privileged state targets, excluding goal-quaternion channels; its reconstruction MSE has weight 0.2. The decoder provides auxiliary supervision and is absent from deployed control. This loss is distinct from the local-controller loss in the main paper.

C.2. Semantic and motor inputs

The motor policy receives 109 target-free proprioceptive/history features, seven measured pose values, and a 768-dimensional goal. Its goal projection has width 64 and a tanh nonlinearity. The inherited recurrent policy emits 21 normalized joint targets. The asset-specific goal quaternion is absent from the migrated student's inference inputs.

The hosted model route recorded in the manifests is `gpt-6-astra`. The local grounding backbone is Qwen3-VL-4B-Instruct. The REVEAL pathway uses public tool actions and a deterministic goal-bank lookup. The fine-tuned pathway predicts a continuous vector within the normalized skill span. Neither pathway receives the requested face's true pose. The local unreadable-view schedule is $(1, 3, 0, 2, 4, 5)$ with 90-step inspections. REVEAL chooses inspection actions of 10–90 steps. This scheduler difference is part of the complete-system comparison and a confound for attributing all effects to model weights.

The language-study archive contains exact task prompts and schedules. The route name alone does not identify immutable serving weights. **[TODO: Reconcile available serving metadata and sampling settings, and include full sanitized system prompts/tool schemas for each release in the anonymous package; state explicitly which provider revisions were not exposed.]**

C.3. Grounding supervision

The grounding model first uses 4,096 teacher-view training examples and 1,024 validation examples, then 4,800 student-view training examples and 2,400 validation examples with mixed replay. Each adaptation stage uses 240 updates. Training and validation collection seeds are disjoint from the formal evaluation seeds. Readability is a geometric/image-signal proxy. The objective combines readability prediction, readable-example skill classification, and cosine alignment to a canonical vector. We do not claim human-verified text recognition labels or an action-consistency objective in this release.

The fixed layout permits learned letter-to-color shortcuts. A held-out pose or trajectory alone does not remove that possibility. A convincing grounding test changes or hides the visual evidence while holding the instruction and physical state fixed, then checks whether the model's selected target changes appropriately.

C.4. Controller features and losses

The sensor branch uses the current proprioceptive vector, measured pose, student belief and proposal, request-anchor

pose and action, fitting residual, binding flag, pending flag, and elapsed pending duration. Its normalized inputs are clipped to $[-8, 8]$. Goal features are centered and scaled using the fixed bank. The sensor, goal, and fusion widths are 192, 64, and 128. Each actor has 22 outputs: a speed logit and 21 residual values.

The waiting actor interpolates from current joint positions toward the proposal and adds a bounded residual. The execution actor applies a bounded residual to the proposal at full speed. Both are supervised from the teacher with residual regularization of 0.01. Waiting labels hold the request anchor. Execution labels follow the active goal. Paired state/goal examples and schedule-derived states are used during fitting. Their causal utility remains to be established by matched training ablations.

No raw RGB, true object pose, teacher action, future model response, or oracle semantic label is a deployed actor input. Vision affects the actor through estimated pose and recurrent belief. The reported stopping gate is measured-pose based. Previously trained completion-evidence heads are inactive and cannot be credited for the current results.

D. Q3: Active Target Search and Visual Feedback

D.1. Q3: Initial planning and continued feedback

The two conditions share 40 initial states. Full feedback reuses Q2's 40 trials from seed 187102, so these trials are not an independent replication. The initial plan uses one observation to return a bounded action sequence. Full feedback uses updated observations and history on up to six calls. Both conditions use the same student, local controllers, stopping rule and time limit.

Protocol	Raw success	Eligible	Drops	Calls
One initial plan	6/40	6/39	0	40
Continued feedback	34/40	33/39	5	120

Table S2. Q3 feedback comparison. One initial-plan HTTP 503 timeout is excluded from both arms for paired analysis. Drops and calls include all scheduled trials.

Twenty-seven pairs succeed only with feedback, and none only with the initial plan (two-sided exact paired $p = 1.49 \times 10^{-8}$). Of 28 eligible initial-plan abstentions, 22 succeed with feedback. Among eligible feedback trials, 36 inspect, 32 first bind after inspection, and 31 successful trials use multiple calls. No trial corrects an executed wrong binding. These counts describe how the agent gathers evidence. They do not tell us whether the target was initially visible or whether every first binding was correct.

Median logical duration is 8.17 seconds for initial planning and 22.0 seconds for feedback on the eligible panel. These durations include unsuccessful termination and must be interpreted alongside 28 initial-plan abstentions. Median measured call latency over raw requests is 8.12 and 5.33 seconds. The protocols differ in output structure and inference budget.

D.2. Q3: Fresh visual evidence

The image intervention uses seeds 187201 and 187202, with 20 trials per seed and arm: 40 matched pairs, 80 trials in total. These are Q3 image-intervention seeds, not additional Q2 controller seeds. Condition order is reversed for the second seed. Both arms freeze the student, predictor, waiting and execution actors, physics, prompt, six-call cap and scoring rule. All 40 pairs are retained, with no service exclusions.

Upper RGB	Success	Drops	Calls	Tokens
Current image	32/40	3	112	209,547
Repeated first frame	9/40	4	98	161,082

Table S3. Q3 image intervention under a common six-call cap. All scheduled trials are included; realized inference cost is not matched.

All 23 pairs with different success outcomes favor current images (two-sided exact paired $p = 2.3842 \times 10^{-7}$, unadjusted). Success counts for the two seeds are $(17, 15)$ with current images and $(5, 4)$ with repeated first frames. Wilson 95% intervals for the combined rates are 65.2–89.5% and 12.3–37.5%, respectively.

Only the upper agent's RGB changes. The frozen arm sends exactly its first frame, including derived crops. Every request also saves the actual current frame. Pixel and file hashes are checked against transport records. All 40 initial states and first images match. Later live frames differ from the first on 72 fresh-arm and 58 frozen-arm requests. Both arms have 31 trials with multiple calls. Local perception, numeric pose feedback, history and motor control remain live in both arms.

Fresh-image outcomes comprise 32 successes, three drops, three false stops and two abstentions. Frozen-image outcomes comprise nine successes, four drops, three false stops and 24 abstentions. Median measured call latency is 6.25 versus 6.95 seconds, and median logical duration is 22.67 versus 25.62 seconds. Actual inference cost and trajectories differ, so this is not a matched-realized-compute test. Timing checks verify continuous logical control and delayed response application, without asserting wall-clock real-time simulation.

A post-hoc analysis checks whether each trial ends with a correct, incorrect or absent target binding. Current images give counts of 34, one and five. Repeated first frames give nine, three and 28. Correct binding alone is insufficient for success: two current-image trials bind correctly but fail the

final pose check. These counts do not measure first-binding accuracy, initial target visibility or stability after stopping. The analysis script records the source hash and each trial's classification.

When correct bindings are acquired. A further post-hoc audit uses applied tool calls rather than terminal states alone. Both arms first declare the correct binding on call one in the same nine trials. With fresh images, an additional 25 trials first bind correctly on calls two through six: eight, nine, five, two and one trials, respectively. Of these 25, 23 subsequently succeed; the other two stop with an incorrect true pose. Repeated first frames produce no additional correct binding after call one. Six fresh-image trials and 31 repeated-frame trials never bind correctly. These are semantic declarations in recorded trajectories, not independently measured first visual-recognition events. All 210 request images were verified against transport hashes. Initial-target readability and finger-versus-back-face occlusion have not been independently annotated and are not reported as results.

D.3. Q3: Adaptive versus fixed inspection

This prospective addendum uses new seeds 187301 and 187302, with 20 trials per seed and arm (40 matched pairs; 80 formal trials). Two admission trials with seed 187291 are excluded from results. Both formal arms were run anew; the earlier fresh-image arm is not reused as a control. The order is adaptive then fixed for the first seed and fixed then adaptive for the second. All 80 outcomes are retained, with no service exclusions, replacement trials or extension after observing the result.

Intervention and controls. Both arms use the archived fresh-image cube release's semantic model (`gpt-6-astra`), system prompt, tool schema, pose estimator, latent student, waiting and execution actors, physical horizon and stopping rule. Each has the same six-call cap. In the adaptive arm, the model selects the color and duration of each `inspect_face` action, from 10 to 90 control steps. In the fixed arm, only these actions are replaced by a cyclic green, yellow, red, blue, magenta, cyan sequence, each lasting 90 steps at 30 Hz. The sweep index advances only on inspection. Non-inspection actions and target declarations are unchanged. Both arms retain the model's choice of whether to inspect, bind, verify, stop or abstain, so this isolates inspection direction/duration rather than removing all high-level autonomy.

The common feedback schema adds the last executed action in both arms, preventing an overridden fixed action from silently being treated as the model's proposed action. Numeric feedback and image history remain live; their realized values and subsequent trajectories differ. Initial physical-state and first-image hashes match for every pair. Runtime contracts, delivered images, applied actions and physical logs passed the final audit. Realized latency and compute are reported rather than assumed equal.

Inspection	Success	Bound	Unbound	Drops	Calls
Fixed sweep	31/40	32	7	1	126
Adaptive	38/40	39	1	0	117

Table S4. New Q3 inspection-policy cohort. Bound counts correct terminal bindings. Fixed inspection additionally has one wrong terminal binding; adaptive has none. Binding categories and terminal failure types overlap and must not be added together.

Successes by seed are $(18, 20)$ for adaptive inspection and $(16, 15)$ for the fixed sweep. Adaptive failures comprise one abstention and one false stop. Fixed failures comprise six abstentions, two false stops and one execution-stage drop. The drop phase describes timing, not its cause. Correct terminal bindings are 39/40 and 32/40, respectively.

Inference and acquisition cost. The prespecified primary endpoint is autonomous success, tested by exact two-sided paired McNemar. The paired table contains 29 shared successes, nine adaptive-only successes, two fixed-only successes and zero shared failures. The observed gain is 17.5 percentage points, but $p = 0.0654297$ does not reject equal success probabilities at $\alpha = 0.05$. Wilson 95% intervals for the individual rates are 83.5–98.6% and 62.5–87.7%; these are not paired-effect intervals. This is one primary test in a new study; binding and cost outcomes are descriptive secondary measures. The result is neither a statistically significant advantage nor evidence of equivalence.

First correct-binding call counts for calls one through six are $(8, 12, 12, 5, 2, 0)$ for adaptive and $(8, 5, 9, 6, 4, 0)$ for fixed inspection. One and eight trials, respectively, never bind correctly. The index includes the input producing the binding; strictly preceding observations equal the index minus one. These never-bound outcomes must accompany any conditional acquisition-time statistic; early failure cannot be interpreted as faster acquisition. The experiment records declared bindings, not an independent visibility annotation.

Adaptive inspection uses 117 calls and 247,383 tokens, versus 126 calls and 273,048 tokens for the fixed sweep. Median measured call latency is 4.57 versus 4.56 seconds; median logical episode duration is 17.83 versus 20.33 seconds, including failures. One prescribed sweep on one cube does not establish superiority to all reactive policies, and an equal maximum call count does not ensure equal realized compute. Different cohorts and hosted-service conditions also preclude attributing differences from older Q3 success rates to this intervention.

D.4. A recorded feedback sequence

Figure S2 shows a recorded pair from the initial-plan/continued-feedback comparison. Both protocols start from the same physical state and byte-identical initial image. The one-call plan abstains. The feedback agent inspects red, then binds T to red after the next observation. The measured gate stops at step 420, with independently scored orientation error 0.0812 radians and correct semantic binding. This is the lowest trial identity among eligible pairs with initial-plan abstention, feedback success, and an inspection action. It is a selected illustration, not a visibility annotation for the whole cohort or evidence of wrong-to-correct binding.

E. Q2: Control During Agent Inference

E.1. Three-seed controller evaluation

The controller ablation uses seeds 187102, 187103 and 187104, with 40 trials per seed and method. Seed 187102 provides the first 40 trials, and the other two seeds add 80. Q2 reports their combined counts as the paper's only formal online controller ablation. For learned slowing alone, we disable joint correction and keep the same initialization, data, sampling, optimizer and 2,400 updates.

Waiting policy	187102	187103 + 187104
Continue	25 / 7 / 5 / 3	48 / 17 / 9 / 6
Fixed slowing	30 / 4 / 0 / 6	58 / 9 / 8 / 5
Learned slowing	28 / 1 / 7 / 4	54 / 6 / 7 / 13
Slowing + correction	34 / 4 / 1 / 1	70 / 1 / 3 / 6

Table S5. Outcome breakdown for the three-seed controller study. The two columns contain 40 and 80 trials per method, respectively. Each cell lists success / waiting drop / execution drop / other failure. Other failures are not further subdivided.

Q1's mapping study and Q3's image intervention use different trials. Their absolute rates need not match Q2, whose controller gains are measured within its own trial set. The initial-plan comparison shares the 40 full-feedback trials from seed 187102, as noted above.

E.2. Waiting-controller fitting

The waiting refit uses training collection seed 185301, validation seed 185302, and fitting seed 185311. These are training/validation seeds, not Q2 evaluation seeds. The inherited actor is updated 2,400 times. Actor batches allocate one half to pending states and one quarter each to unbound and bound execution states. Pending samples are uniform. Only the waiting outputs replace the deployed actor; execution remains unchanged. Formal Q2 evaluation uses the three seeds in Sec. E.1.

E.3. Why speed and joint correction are distinct

A scalar coefficient scales the commanded joint displacement toward the student proposal. It cannot independently change the command direction. Actual motion also depends on the joint servo and contact dynamics. To quantify that restriction, we use 17,755 pending states from 61 validation trajectory groups collected with seed 185302. These states are correlated and contain no formal evaluation trials. For normalized current position q, student proposal u, and the teacher holding action a^*, the best scalar action in hindsight is

$$\alpha^* = \mathrm{clip}\left(\frac{(a^* - q)^\top (u - q)}{\|u - q\|_2^2}, 0, 1\right). \tag{S1}$$

If $u = q$, the interpolation action is q for every coefficient. The resulting error is a lower bound for any scalar-speed predictor on that state. It uses privileged teacher labels for analysis and is not a deployable policy.

Command design	Teacher-action MSE
Continue student	0.5953
Fixed slowdown	0.6095
Learned slowdown	0.4906
Learned slowdown + correction	0.3870
Best scalar in hindsight (bound)	0.4330

Table S6. Held-out waiting-action fit in normalized joint-command coordinates. Lower is better. The privileged bound is distinct from the four deployable conditions. These values are not task success rates.

The full actor is below the scalar bound in 64.8% of these states and has a lower mean error than the bound. This supports the additional action space provided by joint correction. It does not show that every residual is useful or that lower action error necessarily improves closed-loop success. The three-seed Q2 evaluation tests that separate question.

E.4. Timing verification

The request logs verify that pending requests preserve continuous logical control and that replies are applied only after their measured latency and actual completion. Replies arriving after episode termination are not applied. These checks verify the simulated delay protocol rather than wall-clock real-time operation. Q2 aggregate outcomes do not provide per-trial timing for the two additional seeds, so no timing result is extrapolated to all 120 trials.

F. Q1: Cube System Comparisons

These comparisons have their own frozen releases and trial sets. They are not additional Q2 controller seeds. Service-

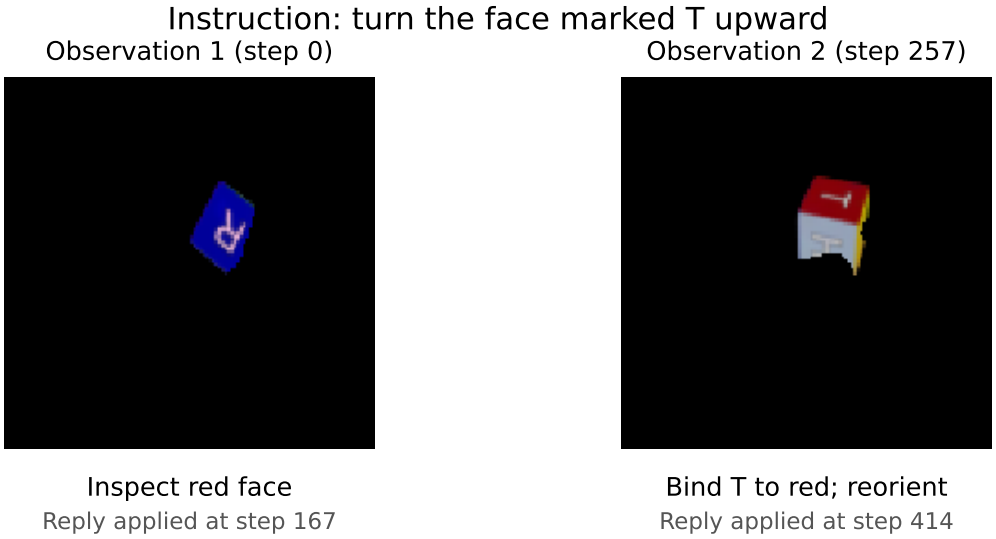


Figure S2. **Inspection followed by grounded execution.** Exact native model-input frames from a completed trial, enlarged without retouching. The requested letter is T. The first observation contains a visible R, and the second exposes T on the red face. Captions summarize recorded public actions. Control continues between request and reply at both observations. The two calls take 5.55 and 5.22 seconds. These are simulator images, not hardware observations.

eligible comparisons remove the same affected slots from all arms; raw records remain archived.

F.1. Completed visual interventions

The upper-level image changes while the low-level pose predictor continues to receive the original image. Lighting is fixed. The permuted condition assigns the next symbol in (T, E, R, L, S, H) to each color face and changes the requested symbol accordingly, preserving the scored physical target across conditions. The masked condition hides the lettering. Geometry-consistent image construction and the transmitted image hashes are audited. Seeds 183441 and 183442 provide 20 trials each, and physical initial-state hashes match across the three conditions.

Method	Identity	Permuted	Masked
REVEAL	32/40	32/40	0/39
Fine-tuned VLM, bare	30/40	16/40	3/40
Rule VLM, bare	23/40	23/40	0/40

Table S7. Completed upper-image interventions. Identity and permutation each use 40 trials. REVEAL masking has one service exclusion. For its paired masking contrast the identity numerator becomes 31/39. All conditions use reconstructed face images, not the untouched native-image comparison.

Fine-tuned success falls by 35 percentage points after remapping and to 7.5% after masking. Rule-based success is 57.5% under both identity and remapping, then zero after masking. Equal aggregate identity and remapped rates do not imply identical trial outcomes. REVEAL now has 32/40 successes under both identity and permutation, with zero of 39 eligible masked trials. This supports robustness to the tested permutation, without proving equivalence or identifying all fine-tuned successes as a fixed lookup.

Applied semantic bindings. We audit every applied, accepted call from the same 240 trials. The episode-file hashes match the archived results, and the last accepted skill agrees with the recorded terminal binding. Table S8 separates acquiring the correct target from completing it. Under permutation, the fine-tuned model first binds correctly in 25 trials and incorrectly in eleven, compared with 34 and four under identity. Nine of its eleven wrong-binding trials select the original color for the newly requested symbol. The rule model has six wrong-binding trials under each layout, with three following the old association after permutation. This pattern is consistent with reliance on training associations, but does not identify the model's internal mechanism.

No wrong-to-correct binding transition occurs in these bare-VLM cohorts. This is descriptive: trajectories, visibility, and opportunities for additional calls differ. Of fourteen fine-tuned false stops under permutation, ten end with a wrong binding and four with the correct binding but an independently incorrect final pose. Masking leads to no accepted binding in 36 fine-tuned trials and all forty rule-model trials.

Model	Layout	Correct	Wrong	None	Old
Fine-tuned	Identity	34	4	2	–
Fine-tuned	Permuted	25	11	4	9
Fine-tuned	Masked	3	1	36	–
Rule	Identity	24	6	10	–
Rule	Permuted	25	6	9	3
Rule	Masked	0	0	40	–

Table S8. First accepted binding in each of 40 trials. Old counts trials with an accepted skill matching the original symbol–color association, defined only for permutation. These are the same trials as the success table, not new evaluations.

The three masked fine-tuned successes cannot establish letter grounding when that information is removed.

Identity versus permutation yields eighteen exclusive successes and four exclusive failures for the fine-tuned model (exact paired $p = 0.00434$), versus ten and ten for the rule model ($p = 1.0$). The final analysis applies Holm correction across six planned image contrasts, giving adjusted $p = 0.0130$ for the fine-tuned mapping contrast. The single deterministic remapping and reconstructed upper images limit generalization claims. REVEAL has five exclusive identity successes and five exclusive permutation successes ($p = 1.0$).

F.2. Native-image comparison and failure audit

Seeds 185451 and 185452 give 40 matched trials per method with no service exclusions. The REVEAL system gives 35 successes, three execution-phase drops, one wrong-binding false stop, and one abstention. The bare fine-tuned system gives 30 successes, nine execution drops, and one horizon failure. The bare rule system gives 17 successes, twelve drops, ten false stops, and one horizon failure. Its drops include two during startup waiting. REVEAL makes 101 calls, fine-tuning 117, and the rule baseline 126. Median latencies are 4.349, 0.099 and 1.104 seconds.

The three REVEAL drops occur during inspection before final target binding. One is 59 control steps after the most recent applied inspection command, and the others are 36 and 24 steps after it. These intervals describe timing, not causal attribution. The false-stop episode binds R to cyan rather than blue. Its measured error to the declared target is 0.057 radians, while true requested-target error is 3.079 radians. It is a semantic binding failure and cannot simply be assigned to pose estimation. The abstention follows four inspections without a declared target. A favorable instantaneous pose during that trajectory does not meet the autonomous success rule.

G. Q1: Native-Layout Dice Extension

The die is a wooden object with one through six native textured pips, rendered at 512×512 with the hand. Three methods each run 40 matched trials over seeds 186301 and 186302. The independent audit verifies all 120 records, canonical targets, paired reset states, response latency and baseline isolation. No trial is service-excluded.

Object-specific motor interface. The released feedforward actor uses ten frames of normalized input. Its goal-related channels are replaced by a learned reconstruction of eight conditioning features. For each frame, a 768-dimensional latent is normalized and encoded to 32 dimensions. The adapter combines it with the four measured quaternion components and their outer product, then applies layers of widths 256, 128, and 8. The original actor and observation normalization remain frozen. Inference receives physical history of shape 10×116 and continuous latent history of shape 10×768. It receives no true target quaternion, target index or table-based reconstruction of its conditioning features.

Perception, control and stop policy. The die uses the released simulator-pose noise, bias, failure, jitter and delay model. There is no image pose estimator in these trials. All upper methods receive the same RGB and image-only crop. The physical horizon is 120 seconds at 20 Hz and the stop gate is checked at 3 Hz. A reply cannot be applied before $\lceil 20L \rceil$ steps, while control continues. The drop boundary and independent full-quaternion success tolerance remain 0.24 metres and 0.16 radians.

REVEAL uses one object-specific maintenance actor during pending and ordinary execution. Its calibrated completion model learns a bounded correction to measured angular error and an uncertainty term. It accepts when the calibrated upper bound is at most 0.16 radians. Once a correct count is declared and the state is not inspection, this gate may stop even with a further request pending. The bare baselines use no learned maintenance or completion module and stop on the measured-angle criterion only when no request is pending. Thus these are complete-system comparisons, not isolated tests of the semantic models or a replication of the cube's shared measured gate.

REVEAL's 95% Wilson interval is [44.6, 73.7]%, versus [35.2, 64.8]% for fine-tuning and [28.5, 57.8]% for rules. REVEAL has ten exclusive successes against six for fine-tuning (exact paired $p = 0.4545$), and eleven against four for rules ($p = 0.1185$). These exploratory comparisons do not establish superiority. REVEAL has twelve startup drops, three execution drops and one false stop. Fine-tuning has eight execution drops, seven false stops and five horizon failures. Rules have seven startup drops, six false stops and

System	Success	%	Drops	Calls
REVEAL	24/40	60.0	15	63
Fine-tuned VLM, bare	20/40	50.0	8	141
Rule VLM, bare	17/40	42.5	7	154

Table S9. Completed native-layout die comparison. No service exclusions. The independent success definition is shared; internal control and stop modules differ by system.

System	Identity	Permuted
REVEAL	28/40	26/40
Fine-tuned VLM, bare	23/40	0/40
Rule VLM, bare	13/40	1/40

Table S10. Dice visual-layout transfer. All 240 scheduled trials are service-eligible. Forty reset states and physical goals are matched across all six cells.

Contrast A versus B	A only	B only	Exact p
I: REVEAL vs. fine-tuned	11	6	0.3323
I: REVEAL vs. rule	18	3	0.0015
P: REVEAL vs. fine-tuned	26	0	2.98×10^{-8}
P: REVEAL vs. rule	25	0	5.96×10^{-8}
REVEAL: I vs. P	4	2	0.6875
Fine-tuned: I vs. P	23	0	2.38×10^{-7}
Rule: I vs. P	13	1	0.0018

Table S11. Exploratory exact two-sided McNemar tests on 40 paired states. Tests are unadjusted. I and P denote identity and permuted layouts.

ten horizon failures. Median request latencies are 5.63, 0.124 and 1.16 seconds, respectively.

H. Q1: Completed Dice Visual-Layout Study

The six cells use identity or a 120-degree visual-mesh rotation about a body diagonal, crossed with three methods. Each cell contains 40 scheduled trials over seeds 186701 and 186702, with matched physical targets and initial states. Collision, inertia and formal solver settings are unchanged. The mesh has slight asymmetries, so this is a visual-layout intervention rather than a pure texture edit. Independent deterministic diagnostics produce identical 32-step state/action traces for native, identity and permuted layers. The formal solver retains its original settings.

Input and motor contract. Printed counts are separated from fixed physical motor IDs. Skills 1–6 display counts $[1, 2, 3, 4, 5, 6]$ in identity and $[2, 4, 6, 1, 3, 5]$ after rotation. This table is used only to construct task instructions and score outcomes. Models receive no layout table, and their commands are not inversely remapped. The learned VLM's continuous 768-dimensional output is applied unchanged. Both layouts share an explicit skill-ID prompt that differs from the native-main prompt, making identity a control for that common change. The VLM retains its native-layout-trained weights. This measures zero-shot layout transfer, not performance after training on permuted examples.

REVEAL receives updated RGB, action/observation history and noisy feedback and can probe skills. The bare VLMs receive only current RGB and instruction, with their original fallback inspection schedule. From a single view, a new printed-count-to-motor-ID association may be ambiguous without a stable face reference. The comparison therefore measures complete systems, not a controlled effect of model scale or fine-tuning alone. Dice maintenance and stopping modules remain as documented above. The shared primary success predicate is unchanged.

Results and paired uncertainty. REVEAL completes 28/40 identity trials and 26/40 permuted trials. Their 95% Wilson intervals are [54.6, 81.9]% and [49.5, 77.9]%. Fine-tuning changes from 23/40 to 0/40, and rules from 13/40 to 1/40. Four paired states succeed only for identity REVEAL and two only for permuted REVEAL. The absence of a detected difference does not establish equivalence or robustness to other layouts. Table S11 reports all seven exploratory paired contrasts without multiple-testing adjustment. The native-main results are a separate cohort and must not be pooled with this study.

Binding and control failures. Permuted REVEAL has 27 trials with a correct binding to a skill ID different from the printed count, and 26 complete successfully. It applies inspection in 32 trials and first binds after inspection in 27. No wrong-to-correct binding transition is observed. Its failures comprise seven startup drops, one execution drop, five abstentions and one wrong-binding stop. These traces describe evidence gathering before commitment, not recovery from an already executed incorrect binding.

The permuted fine-tuned VLM has 31 wrong-binding stops, five horizons and four execution drops. All 31 wrong-binding stops report the native same-number skill. Code review confirms that RGB and instruction enter the backbone and learned latent head, with no service-side assignment from requested count to skill. In identity, it binds correctly in 36 trials but succeeds in 23, illustrating a separate control/stop limitation. The permuted rule model has 32 wrong-binding stops, four horizons, three drops and one success; 24 of its wrong-binding stops select the native same-number skill. These post-hoc diagnostics are consistent with reliance on native associations. They do not establish memorization as an exclusive cause, especially given the baseline's limited



Figure S3. A recorded permuted-layout episode requesting two dots upward. Left: probing physical skill 3 exposes six dots. Right: probing skill 1 exposes two dots. REVEAL then binds the target to skill 1 and stops with an independently measured orientation error of 0.025 radians. Images are the actual model inputs. This first scheduled trial was selected after its outcome and illustrates one successful sequence, not a reliability estimate.

observation history.

Audit and scope. The independent audit checks all 240 trials, 69 frozen source files, matched resets, model-request image hashes, delayed response application, continuous VLM latents and canonical-quaternion scoring. A separate SciPy calculation verifies all six Wilson intervals and seven exact paired tests. No service exclusions, outcome replacements or post-test model changes occur. Trace diagnostics were developed while cells were completing and are explicitly post-hoc. Earlier invalid digit-overlay attempts are excluded and preserved separately. One changed layout, one native-trained local model and object-specific motor weights do not establish broad semantic or geometric generalization.

I. Four Additional Q1 Objects

Duck, hedgehog, pig and milk carton each contribute six cells of forty raw trials, totaling 960. Each object uses its own frozen student and adaptation, native marked RGB for semantic decisions, and noisy simulator pose for motor control. Bare VLM baselines omit the added control modules. We preserve every recorded outcome. Two historical duck service failures and one hedgehog service failure affect only original-mapping cells. Common service-eligible panels remove the corresponding slots from all three methods. All changed-mapping cells retain forty trials.

The full audit covers all 960 episodes, 320 within-mapping initial-image groups and 160 physical-state groups. Seventeen image comparisons differ in one color channel of one pixel by one 8-bit level. Saved model-request pixels and delivered-input hashes match strictly. One duck original-mapping REVEAL reset (slot 7) differs from the other conditions in one quaternion component by 2.98×10^{-8}. The normalized angular difference is 5.77×10^{-8} radians, with identical initial image pixels. We retain this explicitly reviewed numerical exception rather than claiming bitwise reset equality. All other recorded state components match exactly.

As a sensitivity analysis, removing slot 7 from every method gives original-mapping successes of 24/37 for REVEAL, 13/37 for fine-tuning and 15/37 for rules. Removing it from the changed mapping gives 23/39, 2/39 and 9/39. The ordering is unchanged. The main table retains the frozen service-exclusion rule. This post-run sensitivity check does not replace outcomes or change the success definition.

J. Q1: Language and Wording Coverage Across Three Objects

Three separate cohorts contribute 2,160 recorded trials: cube, duck and hedgehog, each with three systems, two languages, two wording templates per language and 60 trials per cell. Cube seeds are 210501–210503, duck seeds 210601–210603, and hedgehog seeds 210701–210703, twenty trials per seed. Each of six goals occurs ten times per cell. These are evaluation blocks, not independent training seeds. Within each object, the same 60 reset seeds and physical goals are reused across methods and instruction variants. Four prompt variants must not be pooled as 240 independent physical tasks.

Intervention and deployment. Mapping, task criterion and object-specific weights remain fixed within each system.

System	English Standard	English Rewrite	Chinese Standard	Chinese Rewrite
Cube: $n = 60, 60, 60, 60$				
REVEAL	42/60	47/60	48/60	36/60
Fine-tuned VLM	42/60	44/60	46/60	47/60
Rule VLM	28/60	27/60	18/60	19/60
Duck: $n = 60, 60, 60, 60$				
REVEAL	41/60	40/60	41/60	46/60
Fine-tuned VLM	23/60	18/60	19/60	21/60
Rule VLM	30/60	15/60	20/60	19/60
Hedgehog: $n = 60, 60, 60, 60$				
REVEAL	44/60	46/60	39/60	46/60
Fine-tuned VLM	23/60	19/60	21/60	15/60
Rule VLM	28/60	23/60	21/60	22/60

Table S12. Raw completion counts over all scheduled trials, retaining infrastructure failures in the denominator. These are separate object-specific cohorts; the four instruction variants reuse physical tasks.

System	English Standard	English Rewrite	Chinese Standard	Chinese Rewrite
Cube				
REVEAL	4	1	3	7
Fine-tuned VLM	0	0	0	0
Rule VLM	0	0	0	0
Duck				
REVEAL	0	1	0	0
Fine-tuned VLM	0	0	0	0
Rule VLM	0	0	0	0
Hedgehog				
REVEAL	0	0	8	0
Fine-tuned VLM	0	0	0	0
Rule VLM	0	0	0	0

Table S13. Service-excluded trial counts, each out of 60 scheduled slots. All raw records are retained.

Cube retains its visual pose estimator, goal bank, latent student, 30-Hz control and measured-pose stopping; REVEAL uses its paper-selected waiting and execution actors. Duck and hedgehog use their own latent students, 20-Hz control and noisy simulator-pose feedback; REVEAL additionally uses object-specific learned maintenance and completion correction. Bare local baselines omit these added actors. All systems use the independently scored 0.16-radian tolerance. Upper RGB is 120-by-120 for cube and 512-by-512 for duck and hedgehog. The English-trained fine-tuned models receive the requested instructions without Chinese retraining. The rule baseline additionally uses localized task/output instructions for Chinese conditions. This is deployed-system language coverage, not an isolated change to the user sentence for every backend, a shared-policy transfer test, or a causal test of fine-tuning. Exact prompts are retained in the evidence bundle.

Service eligibility and raw records. The main-paper language table reports completion conditional on service availability. Within each object and instruction condition, the union of infrastructure-affected slots is removed from all three methods, including any successful baseline outcome at that slot. Thus each row comparison uses identical physical tasks and the same denominator. These rates do not measure end-to-end service reliability. This reporting choice was made after inspecting the raw outcomes; it is not described as preregistered. Table S12 retains every scheduled slot and Table S13 lists exclusions. No excluded slot is replaced; continuations visit only unvisited slots. Physical drops, incorrect stopping, abstention and task-horizon failures remain in the denominator. Infrastructure faults include TLS/proxy failures and server overload, not model grounding or control errors.

Continuation and availability. Service failures interrupted the hedgehog queue. Only unvisited tasks resumed, using separate workers and verified direct HTTPS to the same endpoint after a further SSL failure. Model, weights, prompts, seeds, scoring and one-attempt transport policy remained fixed. All prior outcomes, errors, actual latencies and scheduler changes are retained in the archive.

State pairing and image sensitivity. Cube and duck initial-state hashes match exactly across methods and instruction conditions. Four hedgehog slots differ only in object-quaternion components by at most 5.96×10^{-8}; all other recorded state components match exactly. A post-run numerical reset audit uses a 10^{-6} component tolerance, and the strict sensitivity below removes any nonidentical hashes. Cube first-image hashes match exactly. Some duck and hedgehog renders differ at the pixel level; these do not alter the service-eligible main denominator. Table S14 additionally requires identical initial-state and first-image hashes across all three methods in a condition. All strict paired tests are retained in the data export. The duck differences involve six slots, at most two RGB pixels and one channel level per pixel in 512-by-512 images. Hedgehog image differences involve 12 slots, at most two RGB pixels and one channel level in the same resolution.

Paired inference. Two-sided exact McNemar tests use discordant pairs. Method tests use the same common three-method service-eligible panels as the main table. A language/wording contrast excludes the union of service-affected slots in its two arms; its denominator can differ from a main-table denominator. Holm correction is applied jointly across three objects: one family of 36 language/wording contrasts and a separate family of 36 method contrasts. The strict state-and-image-matched sensitivity applies the same corrections separately. Tables S15 and S16 report all main

System	English Standard	English Rewrite	Chinese Standard	Chinese Rewrite
Cube: $n = 56, 59, 57, 53$				
REVEAL	42/56	47/59	48/57	36/53
Fine-tuned VLM	40/56	43/59	43/57	43/53
Rule VLM	27/56	27/59	18/57	17/53
Duck: $n = 56, 57, 57, 56$				
REVEAL	39/56	40/57	40/57	43/56
Fine-tuned VLM	21/56	17/57	18/57	19/56
Rule VLM	29/56	15/57	20/57	19/56
Hedgehog: $n = 52, 52, 46, 55$				
REVEAL	38/52	40/52	36/46	43/55
Fine-tuned VLM	19/52	15/52	17/46	13/55
Rule VLM	22/52	18/52	16/46	19/55

Table S14. Strict pairing sensitivity: common service-eligible slots additionally require identical initial-state and first-image hashes across all three methods. Numerators are successes. This is separate from the main service-eligible estimand.

System	EN vs. ZH Standard	EN vs. ZH Rewrite	Standard vs. rewrite English	Standard vs. rewrite Chinese
Cube				
REVEAL	1.0000	1.0000	1.0000	0.7637
Fine-tuned VLM	1.0000	1.0000	1.0000	1.0000
Rule VLM	0.4529	1.0000	1.0000	1.0000
Duck				
REVEAL	1.0000	1.0000	1.0000	1.0000
Fine-tuned VLM	1.0000	1.0000	1.0000	1.0000
Rule VLM	1.0000	1.0000	0.0936	1.0000
Hedgehog				
REVEAL	1.0000	1.0000	1.0000	1.0000
Fine-tuned VLM	1.0000	1.0000	1.0000	1.0000
Rule VLM	1.0000	1.0000	1.0000	1.0000

Table S15. Exact paired language/wording tests, Holm-adjusted jointly over 36 contrasts across three objects. Non-significance does not establish equivalence.

adjusted tests. No language/wording contrast survives Holm correction. 20 of 36 method contrasts survive correction. Strict state-and-image-matched sensitivity changes the 0.05 significance decision in 0 of the 72 contrasts; all discordant counts and adjusted values are archived. These tests do not isolate model-only or controller-only effects.

Scope and evidence. Two languages and two templates per language, with six fixed targets and one mapping per object, do not establish broad multilingual generalization or equivalence. Earlier misconfigured runs are excluded; historical native/mapping cohorts remain separate. The sanitized archive retains all 2,160 outcomes, 72 main and 72 sensitivity comparisons, prompts, schedules and checksums.

Condition	REVEAL–FT	REVEAL–Rule	FT–Rule
Cube			
EN standard	1.0000	0.1220	0.1487
EN rewrite	1.0000	0.0102	0.0424
ZH standard	1.0000	< 0.0001	< 0.0001
ZH rewrite	1.0000	0.0017	< 0.0001
Duck			
EN standard	0.0258	0.3458	1.0000
EN rewrite	0.0043	0.0011	1.0000
ZH standard	0.0043	0.0102	1.0000
ZH rewrite	< 0.0001	0.0001	1.0000
Hedgehog			
EN standard	0.0025	0.1120	1.0000
EN rewrite	< 0.0001	0.0004	1.0000
ZH standard	0.0013	0.0041	1.0000
ZH rewrite	< 0.0001	0.0011	1.0000

Table S16. Exact paired method tests on the main common service-eligible panels, Holm-adjusted over a separate family of 36 contrasts. FT denotes fine-tuning. Strict pairing sensitivity tests are archived.

K. Independent Pose Estimation under Hand Occlusion

This supporting perception study for Q4 evaluates fixed RGB and RGB-D estimators on a 58.5 mm lettered cube. Its RGB model differs from the frozen agent CNN, and its pose recalls do not update agent success counts.

Sampling and scoring. Three test seeds (31001–31003), four camera pitches (0°, 60°, 90°, 105°), and 512 frames per group yield 6,144 observations in 48 environment clusters. RGB, RGB-D, their no-hand controls and perturbed-depth RGB-D produce 30,720 predictions. Collection records four geometry rejections; 6,139 admitted observations contain the full object.

Recall requires translation error < 10 mm and full rotation error $< 10°$, without cube-symmetry minimization. Invalid predictions count as failures. Secondary recall uses 5 mm and 5°. ADD uses 6,144 nominal-cube surface points and threshold 0.1 diameter, not BOP average recall. Frozen prose used non-strict inequalities, but all evaluations consistently used strict thresholds. No samples lie on the primary boundaries.

Accuracy and uncertainty. RGB passes on 2,306/6,144 observations and RGB-D on 3,977/6,144, a difference of 27.20 percentage points. The reported 95% intervals are [36.60,38.48]% for RGB, [64.02,65.46]% for RGB-D, and [26.07,28.30] points for the paired difference. Intervals use 10,000 paired bootstrap draws of whole environment clusters, stratified by seed, retaining their views and temporal observations together. They describe sampling variation for this

Hand occlusion	Frames	RGB	RGB-D
[0,25%)	2,891	66.17	97.61
[25,50%)	1,964	19.91	55.45
[50,75%)	1,198	0.17	5.51
[75,100%]	91	0.00	0.00
All	6,144	37.53	64.73

Table S17. Independent pose recall (%) under hand occlusion. All frames remain in the denominator, including invalid predictions. Occlusion is associated with viewpoint in this sampling design.

Metric	RGB	RGB-D
Recall, 10 mm / 10 deg. (%)	37.53	64.73
Recall, 5 mm / 5 deg. (%)	14.63	47.56
ADD recall (%)	38.35	71.48
Translation mean (mm)*	57.56	6.26
Rotation mean (deg.)*	12.28	10.82
Invalid predictions	69/6144	1/6144

Table S18. Secondary pose metrics. All recall denominators include invalid predictions. *Mean errors use valid predictions only and must be interpreted with the invalid counts. Threshold comparisons are strict.

object and these test seeds, not variation across trained models or unseen objects. Table S17 shows that gains concentrate in light and moderate occlusion. At 50–75%, RGB-D recall is only 5.51%, and both systems fail all 91 observations at $\geq 75\%$.

Paired controls and sensitivity. Removing hand occlusion at matched poses gives recall of 76.90% for RGB and 99.89% for RGB-D. The overall occlusion-associated loss is therefore 39.37 versus 35.16 points, with a paired reduction of 4.22 points [2.77,5.62]. This reduction is not uniform: at 50–75% occlusion, the loss is 72.20 versus 94.49 points. The higher unoccluded RGB-D ceiling does not imply robustness when most of the object is hidden. The no-hand RGB render retains soft Gaussian-splat boundaries, whereas the occluded image uses a hard visible-object mask, which also limits this control. Restricting to fully framed observations gives 37.56% versus 64.78%, similar to the full cohort.

Adding 1 mm Gaussian ray-depth noise and dropping 20% of points yields RGB-D recall of 3,973/6,144 (64.66%). This is a sensitivity check, not a calibrated physical depth-camera error model. Per-seed RGB/RGB-D recall is 37.30/64.89%, 37.70/64.36%, and 37.60/64.94%. Viewpoint and occlusion covary: recall at nominal view is 76.04/99.80%, versus 0.85/4.23% at 105°.

Training and interpretation. The RGB system uses development-selected letterboxing and its existing PnP decoder. RGB-D uses the supplied epoch-1812 weights. Training histories and camera adaptation differ, so this is not a matched-budget architecture or depth-input ablation. RGB-D training manifests record 20,000 samples from seeds 43–50, validation uses seed 2027, and the test seeds do not overlap. Only nine training entries have more than 50% occlusion. This post-hoc manifest observation is consistent with the severe-occlusion failure but is not causal evidence, and the full training imagery has not been re-audited. The old RGB training-data lineage is incomplete.

The evaluation uses simulator pose labels, ideal target segmentation, and virtual co-located depth. Real-camera calibration and Gaussian-splat appearance assets are part of the setup, so it should not be described as using no real-world information. The evidence supports improved simulated pose accuracy over this RGB deployment, with a clear severe-occlusion limit. It does not yet establish improved agent completion or real-camera accuracy. The earlier 39/40 original pose-goal versus 35/40 latent-student physical-reach diagnostic used the frozen agent CNN and is a separate experiment.

Evidence provenance. The supplied integrity report records 129 unchanged frozen files, shared finite ground truth, and pose-error recomputation within 2.28×10^{-13} mm and 1.01×10^{-11} degrees. Paper integration verifies file hashes and re-sums twelve shard counts. Bootstrap intervals and per-frame checks retain delivery provenance. Protocol/model hashes are archived; the final reproduction package still needs sanitized per-frame records.

L. Physical-Hand Evaluation and Remaining Scope

Physical-hand results are not yet available. The planned evaluation will identify the tested objects, camera calibration, reset procedure and independent scorer, and record synchronized images and commands. The simulation evidence measures terminal binding and orientation. Direct visibility and post-stop holding should be evaluated separately to test sustained target presentation. The following layouts reserve space for the planned hardware evidence.

M. Reproducibility

The archived evidence retains outcomes, goals, stopping errors, service exclusions and latency for the audited cohorts. The three-seed controller aggregate is maintained separately from the historical trial records.

[TODO: Complete anonymous code and asset licenses.]

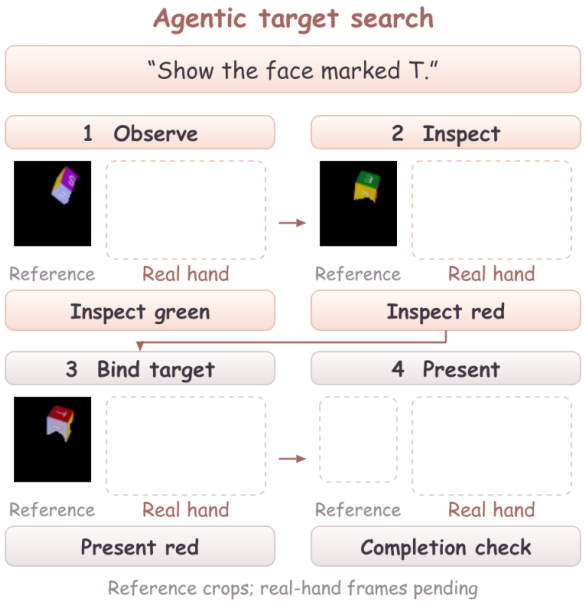


Figure S4. **Planned inspection and presentation sequence.** Reference crops illustrate observing, inspecting, binding, and presenting a requested face. Real-hand image slots remain empty. This is a protocol mockup, not a physical-robot result.

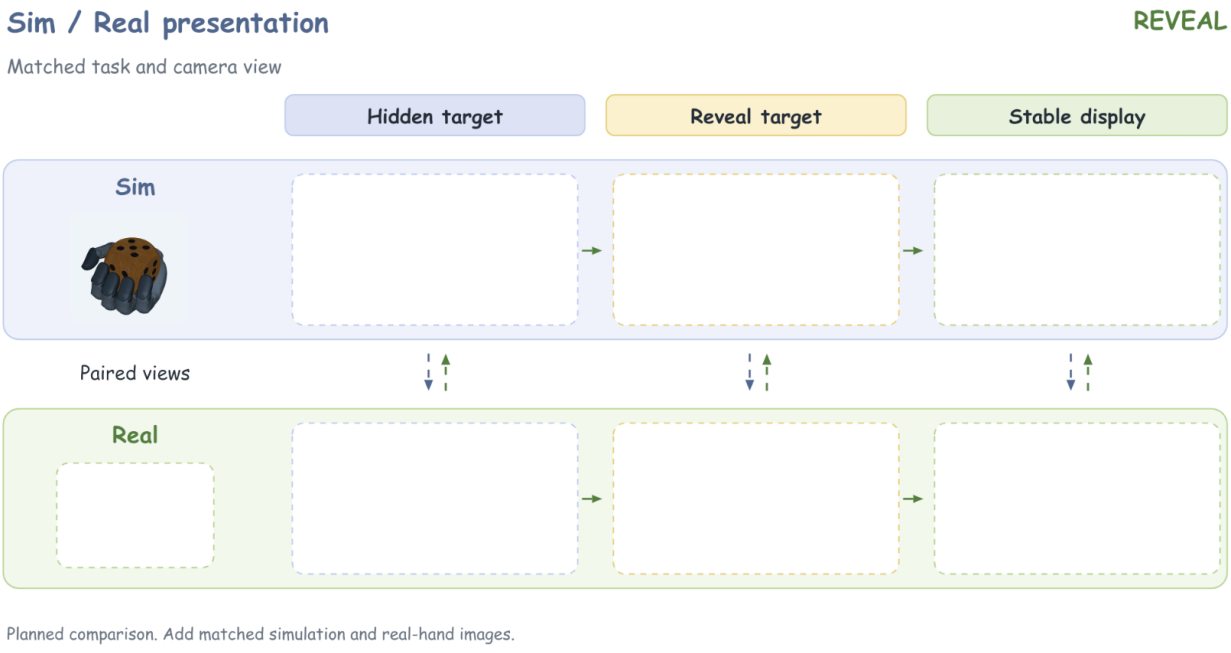


Figure S5. **Planned simulation/physical presentation comparison.** The layout pairs hidden-target, reveal, and stable-display stages. The reference simulation crop is illustrative; the remaining simulation and physical-hand slots await matched observations. No real-robot outcome is represented.

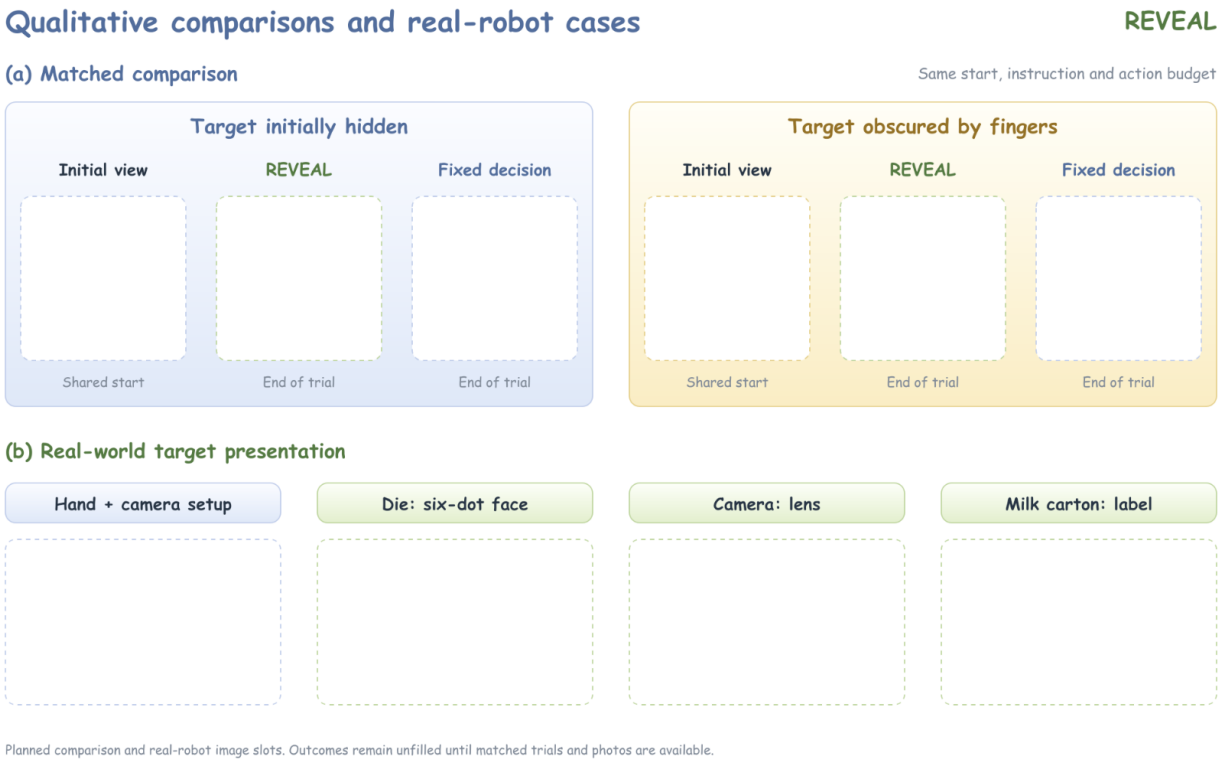


Figure S6. **Planned qualitative comparison and real-robot cases.** Initial/end-state and physical-robot image slots remain placeholders until corresponding observations are available. No completed trial is represented.

References

[1] Arjun Bhardwaj, Maximum Wilder-Smith, Mayank Mittal, Vaishakh Patil, and Marco Hutter. ViserDex: Visual sim-to-real for robust dexterous in-hand reorientation. *arXiv preprint arXiv:2604.11138*, 2026.